

Engineering Mathematics: Introduction to Partial Differentiation

Partial differentiation is a powerful mathematical tool used to analyze functions of multiple independent variables. Unlike ordinary differentiation, where a function depends on a single variable, partial differentiation treats all variables as constants except for the one with respect to which the differentiation is being performed.

Core Concept: The Fundamental Rule

When finding the partial derivative of a function $f(x, y, z, \dots)$ with respect to an independent variable x , denoted as $\frac{\partial f}{\partial x}$:

- Treat x as the **only** variable.
- Treat $y, z, \dots$ as **pure constants**.
- Apply the standard rules of single-variable calculus (Chain Rule, Product Rule, Quotient Rule).

Worked Examples

Problem 1: Logarithmic and Trigonometric Composition

Given the function:

$$u = \log(\tan x + \tan y)$$

Find the partial derivative with respect to x , denoted as $\frac{\partial u}{\partial x}$.

Step-by-Step Solution

Apply the Chain Rule for differentiation, treating y (and consequently $\tan y$) as a constant:

$$\frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y} \times \frac{\partial}{\partial x}(\tan x + \tan y)$$

Because $\tan y$ is treated as a constant, its derivative with respect to x is 0, and the derivative of $\tan x$ is $\sec^2 x$:

$$\frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y} \times (\sec^2 x + 0)$$

Simplify the expression:

$$\frac{\partial u}{\partial x} = \frac{\sec^2 x}{\tan x + \tan y}$$

Problem 2: Variable Base vs. Variable Exponent

Given the function:

$$z = x^y$$

Find the partial derivatives $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Case 1: Differentiating with respect to x

Treat y as a constant exponent. This matches the standard power rule form $\frac{d}{dx}(x^n) = nx^{n-1}$:

$$\frac{\partial z}{\partial x} = yx^{y-1}$$

Case 2: Differentiating with respect to y

Treat x as a constant base. This matches the standard exponential rule form $\frac{d}{dy}(a^y) = a^y \log a$:

$$\frac{\partial z}{\partial y} = x^y \log x$$

Problem 3: Exponential Composition

Given the function:

$$z = e^{x+y}$$

Find the partial derivative with respect to x , denoted as $\frac{\partial z}{\partial x}$.

Step-by-Step Solution

Apply the Chain Rule, treating y as a constant:

$$\frac{\partial z}{\partial x} = e^{x+y} \times \frac{\partial}{\partial x}(x+y)$$

Evaluate the inner derivative:

$$\frac{\partial z}{\partial x} = e^{x+y} \times (1+0)$$

Simplify the expression:

$$\frac{\partial z}{\partial x} = e^{x+y}$$

Problem 4: Radical Composition inside a Trigonometric Function

Given the function:

$$u = \cos(\sqrt{x} + \sqrt{y})$$

Find the partial derivative with respect to x , denoted as $\frac{\partial u}{\partial x}$.

Step-by-Step Solution

Apply the Chain Rule, treating y (and $\sqrt{y}$) as a constant:

$$\frac{\partial u}{\partial x} = -\sin(\sqrt{x} + \sqrt{y}) \times \frac{\partial}{\partial x}(\sqrt{x} + \sqrt{y})$$

Evaluate the inner partial derivative using the rule $\frac{d}{dx}(x^{1/2}) = \frac{1}{2\sqrt{x}}$:

$$\frac{\partial u}{\partial x} = -\sin(\sqrt{x} + \sqrt{y}) \times \left(\frac{1}{2\sqrt{x}} + 0 \right)$$

Simplify the final product:

$$\frac{\partial u}{\partial x} = -\frac{1}{2\sqrt{x}} \sin(\sqrt{x} + \sqrt{y})$$

Problem 5: Missing Term Fill-in (Three Variables)

Given the incomplete derivation step for $u = e^{x^2+y^2+z^2}$ to find $\frac{\partial u}{\partial z}$:

$$\begin{aligned}\frac{\partial u}{\partial z} &= e^{x^2+y^2+z^2} \times \frac{\partial}{\partial z}(x^2 + y^2 + z^2) \\ \frac{\partial u}{\partial z} &= e^{x^2+y^2+z^2} \times (2z + \dots)\end{aligned}$$

Complete Analysis

Treat x and y as constants. Therefore, $\frac{\partial}{\partial z}(x^2) = 0$ and $\frac{\partial}{\partial z}(y^2) = 0$.

The complete structural step evaluates to:

$$\begin{aligned}\frac{\partial u}{\partial z} &= e^{x^2+y^2+z^2} \times (2z + 0 + 0) \\ \frac{\partial u}{\partial z} &= 2ze^{x^2+y^2+z^2}\end{aligned}$$

Problem 6: Polynomial Terms

Given the function:

$$u = x^2y^3 + x^3y^2$$

Find the partial derivative with respect to x , denoted as $\frac{\partial u}{\partial x}$.

Step-by-Step Solution

Apply the sum rule and factor out the constant y terms from each differential term:

$$\frac{\partial u}{\partial x} = y^3 \frac{\partial}{\partial x}(x^2) + y^2 \frac{\partial}{\partial x}(x^3)$$

Compute the derivatives using the power rule:

$$\frac{\partial u}{\partial x} = y^3(2x) + y^2(3x^2)$$

Rearrange terms into standard algebraic notation:

$$\frac{\partial u}{\partial x} = 2xy^3 + 3x^2y^2$$

Problem 7: Product of Independent Functions

Given the function:

$$u = 10^x \cos y$$

Find the partial derivative with respect to x , denoted as $\frac{\partial u}{\partial x}$.

Step-by-Step Solution

Because $\cos y$ acts entirely as a constant coefficient when differentiating with respect to x , pull it out of the derivative:

$$\frac{\partial u}{\partial x} = \cos y \times \frac{\partial}{\partial x}(10^x)$$

Apply the exponential rule $\frac{d}{dx}(a^x) = a^x \log a$:

$$\frac{\partial u}{\partial x} = \cos y \times (10^x \log 10)$$

Reorder the factors:

$$\frac{\partial u}{\partial x} = 10^x \log 10 \cdot \cos y$$

Problem 1: Constant Coefficients and Product Terms

Given the function:

$$z = ax^2 + by^2 + 2abxy$$

Find the partial derivative with respect to x , denoted as $\frac{\partial z}{\partial x}$.

Step-by-Step Solution

Differentiate each term sequentially with respect to x , treating a , b , and y as constants:

For the first term ax^2 , apply the standard power rule:

$$\frac{\partial}{\partial x}(ax^2) = 2ax$$

For the second term by^2 , because there is no x component present, the entire term behaves as a standalone constant:

$$\frac{\partial}{\partial x}(by^2) = 0$$

For the third term $2abxy$, treat $2aby$ as a collective constant multiplier for x :

$$\frac{\partial}{\partial x}(2abxy) = 2aby \cdot \frac{\partial}{\partial x}(x) = 2aby(1)$$

Combine the evaluations to form the complete derivative:

$$\frac{\partial z}{\partial x} = 2ax + 0 + 2aby$$

$$\frac{\partial z}{\partial x} = 2(ax + aby)$$

Problem 2: Quotient Rule Optimization

Given the implicit equation:

$$z(x + y) = x - y$$

Find the value of the expression:

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} \right)^2$$

Step-by-Step Solution

Step 2.1: Isolate the Dependent Variable

Rearrange the equation to express z explicitly as a function of x and y :

$$z = \frac{x - y}{x + y}$$

This sets up a quotient form $\frac{u}{v}$. Recall the Quotient Rule for differentiation: $\frac{v \cdot u' - u \cdot v'}{v^2}$.

Step 2.2: Compute $\frac{\partial z}{\partial x}$

Treat y as a constant. Differentiate the numerator and denominator with respect to x :

$$\frac{\partial z}{\partial x} = \frac{(x + y) \cdot \frac{\partial}{\partial x}(x - y) - (x - y) \cdot \frac{\partial}{\partial x}(x + y)}{(x + y)^2}$$

$$\frac{\partial z}{\partial x} = \frac{(x + y)(1 - 0) - (x - y)(1 + 0)}{(x + y)^2}$$

$$\frac{\partial z}{\partial x} = \frac{x + y - x + y}{(x + y)^2}$$

$$\frac{\partial z}{\partial x} = \frac{2y}{(x + y)^2}$$

Step 2.3: Compute $\frac{\partial z}{\partial y}$

Treat x as a constant. Differentiate the numerator and denominator with respect to y :

$$\frac{\partial z}{\partial y} = \frac{(x + y) \cdot \frac{\partial}{\partial y}(x - y) - (x - y) \cdot \frac{\partial}{\partial y}(x + y)}{(x + y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{(x + y)(0 - 1) - (x - y)(0 + 1)}{(x + y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{-(x + y) - (x - y)}{(x + y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{-x - y - x + y}{(x + y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{-2x}{(x + y)^2}$$

Step 2.4: Substitute into the Target Expression

Evaluate the difference internal to the bracket:

$$\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = \frac{2y}{(x + y)^2} - \left(\frac{-2x}{(x + y)^2} \right)$$

$$\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = \frac{2y + 2x}{(x + y)^2}$$

Factor out 2 from the numerator to enable cancellation:

$$\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = \frac{2(x + y)}{(x + y)^2}$$

$$\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = \frac{2}{x + y}$$

Square the resulting fraction to complete the problem requirements:

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} \right)^2 = \left(\frac{2}{x + y} \right)^2$$

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} \right)^2 = \frac{4}{(x + y)^2}$$